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MANAGEMENT REVIEW

Management without theatre.



OPERATIONS

PROBLEM SOLVING WITH A3 AND PARETO

Why problems that are worked around come back, what the Pareto principle is and is not, how Toyota puts a problem on one sheet, and an A3 card to fill in.

239 h of watching nurses in nine hospitals: most problems were worked around rather than fixed at the cause.

TUCKER & EDMONDSON, 2003

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Every operation has problems that return each week: the same late start, the same missing part, the same complaint. Getting through the shift is one thing. Making sure the problem stays away is another, and that is the subject of these pages.

Researchers who followed nurses in nine hospitals found that most problems were worked around and few reached someone who could remove the cause. The Pareto chart shows where to start; Joseph Juran gave the idea Pareto's name and later admitted it was the wrong one. Toyota puts a whole problem, from the background to the follow-up, on one sheet of A3 paper. A study of 18 hospital projects suggests that following the steps pays off.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Worked around, not solved

Anita Tucker and Amy Edmondson followed nurses in nine hospitals as they ran into failures in their daily work, and recorded what they did next.

239 h	26	9
of observation on the wards	nurses followed through their shifts	hospitals

First-order problem solving

The nurse finds a way around the problem

The patient is cared for, the shift goes on

The cause stays where it was

Second-order problem solving

The nurse deals with the immediate problem

and tells someone who can change the cause

The next shift may not meet it again

Working around was the dominant response. The nurses were not careless: under time pressure they got the job done, and in doing so kept the problem out of sight of the people who could have removed it.

OUR READING

A workaround gets one shift through the problem. It also hides the problem from everyone who could stop it coming back.

Source: Anita L. Tucker & Amy C. Edmondson, California Management Review, 2003

THE PRINCIPLE

A principle with the wrong name

In the mid-1920s a young engineer, Joseph Juran, noticed that quality defects are not equally frequent: a few kinds make up most of them. Years later he gave the idea the name of the Italian economist Vilfredo Pareto.

1896
Pareto publishes his work on the unequal distribution of incomes.

1920s
Juran sees that a few kinds of defects make up most of them.

1974
Juran's “Mea culpa”: the name was a mistake. It stays anyway.

“I was forced to confess that I had mistakenly applied the wrong name to the principle.”

That is how Juran put it in 1974. He had called it the principle of “the vital few and the trivial many”. Pareto had written about wealth and incomes, not defects; the step to quality was Juran's own. Later he preferred “the vital few and the useful many”, so that the rest would not be read as worthless.

OUR READING

80/20 is a rule of thumb: a reminder to count before you decide where to start.

Source: Joseph M. Juran, Juran Institute, 1974

Juran's paper is dated 1974 in the copy of the Juran Institute; a bibliography lists it in Quality Progress in May 1975.

More in the essay Pareto and 5 Why in practice, not in PowerPoint

THE MODEL

One problem, one sheet

The A3 report is a practice Toyota pioneered: the problem, the analysis, the countermeasures and the plan on a single sheet of A3 paper, 297 × 420 mm. John Shook calls it standardised storytelling.

01	**Background**	Why this problem matters now.
02	**Current condition**	What happens today, drawn and measured.
03	**Goal**	What will be different, by how much, by when.
04	**Analysis**	Why it happens: the cause behind the cause.
05	**Countermeasures**	What will change, aimed at the cause rather than the symptom.
06	**Plan**	Who does what, by when.
07	**Follow-up**	How the result is checked, and what happens if it does not come.

Shook writes that managers use the A3 to teach others to find root causes and to think with evidence, and to bring people and departments to agreement.

OUR READING

The sheet is small on purpose. A problem that will not fit on one page has not yet been understood.

Sources: Lean Enterprise Institute, 2014; John Shook, MIT Sloan Management Review, 2009 (via Opening of the article; the rest in excerpts)

The sections follow Shook and the Lean Enterprise Institute; the descriptions are the editors' own.

HOW IT IS APPLIED

Eight steps and a check

In 2004 Toyota gathered its ways of solving problems into the Toyota Business Practices: eight steps, from describing the problem to making the solution the new standard.

Plan

1 Clarify the problem

2 Break it down

3 Set a target

4 Find the root cause

5 Develop countermeasures

Do, check, act

6 See the countermeasures through

7 Check the results and the process

8 Make what worked the standard

A study by Ghosh and Sobek looked at 18 improvement projects in one hospital. The projects that followed every step of the problem-solving routine improved more. One hospital and a correlation are a hint; they prove nothing.

OUR READING

Many attempts fail at step one: the problem is written down as a solution, “we need more people”, instead of as the gap between what should happen and what does.

Sources: Toyota Motor Corporation, 2012 (via Toyota's history page; the steps from summaries); M. Ghosh & D. K. Sobek II, Journal of Health Organization and Management, 2015 (via Abstract)

The steps are Toyota's, in our words; the grouping into plan, do, check and act follows summaries of the method.

HOW IT IS MEASURED

A Pareto chart in four steps

As ASQ describes it, the chart sorts the bars from longest to shortest and adds them up with a line.

01	**Choose one measure**	A count or a cost, over one period.
02	**Group by cause**	Categories you can act on; “other” goes last.
03	**Sort and add up**	Longest bar first, then the running percentage. Start on the left.
04	**Draw it again**	After the change, for the same period and the same categories.

REPORTED INJURIES TO EMPLOYEES BY KIND, GREAT BRITAIN, 2024/25



Once sorted, the first two kinds account for 47% of the injuries employers reported, the first five for 75%. A safety A3 would begin with the first two bars.

Sources: ASQ, 2026 (via Excerpts, through web search); Health and Safety Executive, 2025

Data: HSE, injuries employers must report (RIDDOR). The sums and the steps are the editors'.

Open the tool: Pareto 80/20 stivencatalyst.com/tools/pareto/

TOOL OF THE ISSUE

The A3 card

One sheet for one problem. Fill it in with the people who do the work, and in pencil: the sheet changes as you learn.

01 The problem and its owner in a few words, one name, a date

02 Background why it matters now

03 Current condition what happens today, with a number

04 Goal what changes, by how much, by when

05 Root cause why, and why again, until you reach something you can change

06 Countermeasures and plan what, who, by when

07 Follow-up when you check, and what you do if the number does not move

Source: Lean Enterprise Institute, 2014

Proposed by the editors; the sections follow the Lean Enterprise Institute and John Shook.

Open the tool: 5 Whys stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Why Hospitals Don't Learn from Failures
Anita L. Tucker & Amy C. Edmondson, California Management Review, 2003
https://cmr.berkeley.edu/2003/02/45-2-why-hospitals-dont-learn-from-failures-organizational-and-psychological-dynamics-that-inhibit-system-change

The Non-Pareto Principle; Mea Culpa
Joseph M. Juran, Juran Institute, 1974
https://www.juran.com/wp-content/uploads/2021/03/The-Non-Pareto-Principle-1974.pdf

Lean Lexicon: A3 report
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/a3-report/

Toyota's Secret: The A3 Report
John Shook, MIT Sloan Management Review, 2009 · via Opening of the article; the rest in excerpts
https://sloanreview.mit.edu/article/toyotas-secret-the-a3-report/

75 Years of Toyota: Toyota Business Practices
Toyota Motor Corporation, 2012 · via Toyota's history page; the steps from summaries
https://www.toyota-global.com/company/history_of_toyota/75years/text/leaping_forward_as_a_global_corporation/chapter4/section7/item4_a.html

A problem-solving routine for improving hospital operations
M. Ghosh & D. K. Sobek II, Journal of Health Organization and Management, 2015 · via Abstract
https://doi.org/10.1108/JHOM-09-2013-0191

What is a Pareto Chart?
ASQ, 2026 · via Excerpts, through web search
https://asq.org/quality-resources/pareto

Kind of accident statistics in Great Britain, 2025
Health and Safety Executive, 2025
https://www.hse.gov.uk/statistics/assets/docs/kinds-of-accident.pdf

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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